skkmigas PERTAMINA EP	**PERTAMINA EP REGIONAL 1 ZONA 4**	POG
ELECTRICAL CABLE SIZING CALCULATION	No. Dok. : TMBP-SP001-EL-CAL-D103 Rev. : A Halaman : 1 dari 21	

NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANGAN TIMUR - FIELD LIMAU**

NO. KONTRAK : **SA02035383B**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	09/01/2025	Issued For Review	FN	AS	RTN		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

TABEL REVISI

Rev.	Tanggal	Deskripsi

COMMENT RESPONSE SHEET

No.	Page	Company's Comment	Contractor's Response	Remarks

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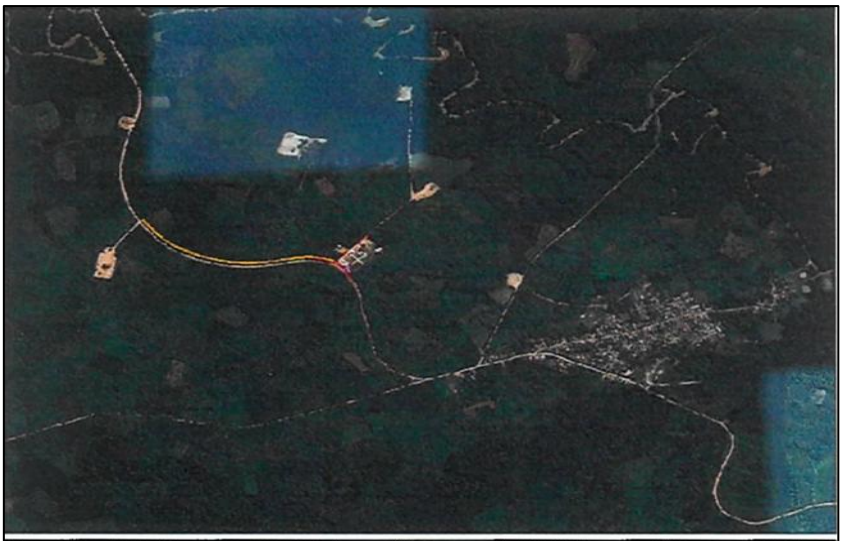
1. GENERAL

1.1 Latar Belakang

Struktur Tanjung Miring Barat (TMB) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 20 kilometer di sebelah tenggara Kota Prabumulih, dengan luas daerah sekitar 8 km². Struktur Karangan (KRG) terletak di Kabupaten Muara Enim dan Kota Prabumulih, sekitar 15 kilometer di sebelah selatan Kota Prabumulih, dengan luas daerah sekitar 45 km².



ID	Easting (Meter)	Northing (Meter)	Elevation (Meter)	Feature Code
PN-1531	424962,745	96 7376,210	10,414	
PM-8593	425000,093	907528.777	19,046	

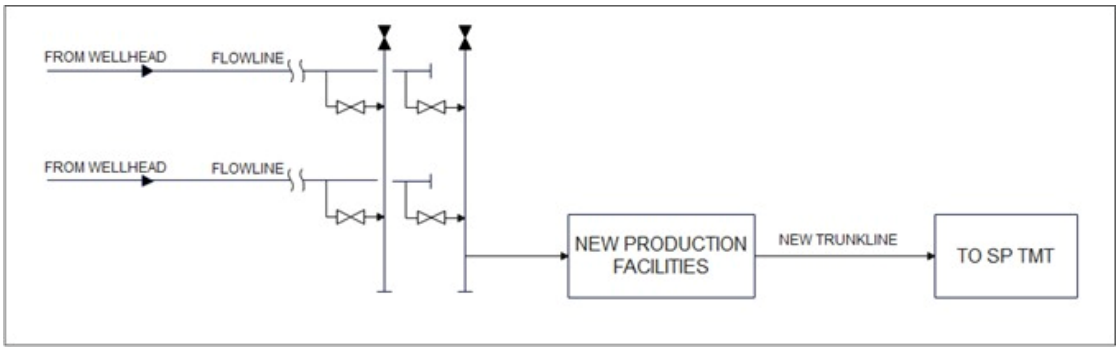
Gambar 1. 1 Lokasi Rencana SP Tanjung Miring Barat – Karangan Timur

Lapangan Tanjung Miring Barat telah diproduksi sejak tahun 1938 dan terus beroperasi hingga tahun 2003 dengan produksi terakhir dari sumur TMB-11. Sejak dinonaktifkannya lapangan ini, fasilitas produksi yang terdiri dari flowline, stasiun pengumpul, serta jalur pengiriman pipa minyak ke custody transfer (PPP Prabumulih) sudah tidak layak operasi. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Baral serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah. Saat ini status SP TMB adalah idle atau tidak beroperasi, sehingga produksi sumuran menggunakan Tank On-Site (TOS) dan dikirim secara gross ke SP-II TMT menggunakan trucking. Dikarenakan tidak adanya line gas ke sumur-sumur struktur TMB, supply Listrik untuk pumping unit menggunakan solar (diesel engine). Lapangan Karangan Timur hanya diproduksikan dari 2 sumur yaitu KRG-9 dan KRG-10 beroperasi mulai tahun 1984, dimana kondisi eksisting jalur pengiriman minyak dari sumur menggunakan trucking. Apabila akan dilakukan pengembangan lapangan, diperlukan pembangunan fasilitas produksi baru yang dapat menampung produksi hidrokarbon dari lapangan Tanjung Miring Barat serta memprosesnya agar sesuai dengan spesifikasi yang dipersyaratkan di titik serah.

Dalam rencana jangka panjang sebagai upaya peningkatan produksi, PT Pertamina EP berencana akan mengembangkan Lapangan Tanjung Miring Barat dan Karangan Timur melalui

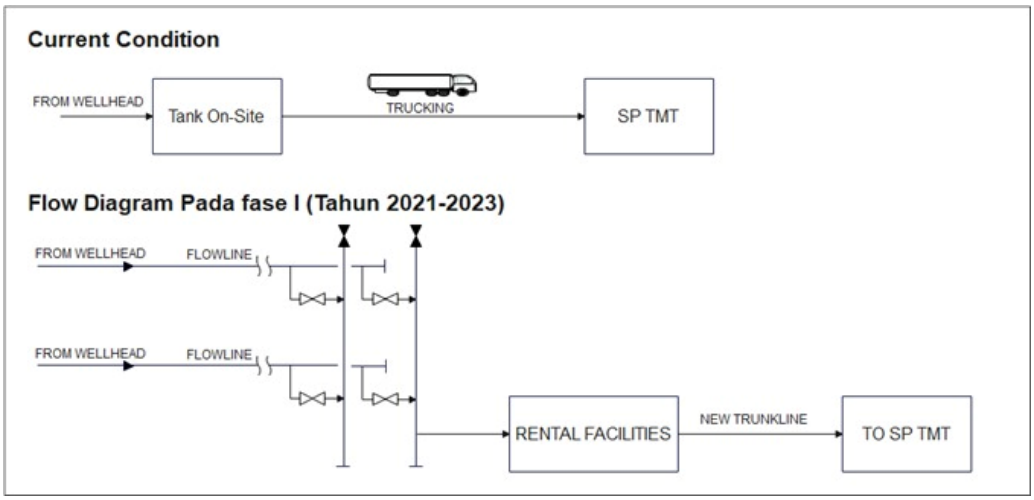
pengembangan fasilitas produksi dimana rencana kerja pengembangan fasilitas akan dibagi ke dalam dua fase.

Pada fase pertama, produksi dari pengembangan awal TMB -KRG Timur akan dikirimkan menuju fasilitas sewa Early Production Facility (EPF) untuk selanjutnya dikirimkan menuju SP- 1 TLJ melalui trunkline baru. Pada fase ini lingkup fasilitas produksi yang akan dibangun adalah flow/ine dari sumuran, sewa EPF dan trunkline dari EPF sampai dengan SP-1 TLJ.



Gambar 1. 2 Rencana Pola Operasi Fase

Pada fase kedua, akan dilakukan pengembangan tambahan dari sumur-sumur TMB-KRG Timur. Pada fase ini, lingkup fasilitas produksi yang akan dibangun meliputi pembangunan flowline dari sumuran pengembangan lanjutan serta peningkatan fasilitas SP TMB untuk menggantikan sewa EPF.



Gambar 1. 3 Rencana Pola Operasi Fase

1.2 Tujuan

Tujuan dari dokumen ini adalah untuk memberikan informasi terkait perhitungan ukuran kabel (cable sizing) dan penurunan tegangan (voltage drop) yang terjadi dalam sistem. Perhitungan ini dilakukan sebagai bagian *dari project Engineering Design & Fabrication of Flow Loop Demo Unit*. Dokumen ini juga akan menjadi basis pemilihan kabel pada saat proses procurement.

1.3 Ruang Lingkup

Ruang lingkup dokumen *Electrical Cable Sizing* ini mencakup kegiatan perhitungan dan penentuan ukuran kabel listrik pada sistem tegangan rendah (Low Voltage) untuk proyek *Engineering Design & Fabrication of Flow Loop Demo Unit*, yang dilakukan pada tahap *Detailed Engineering Design*. Perhitungan meliputi analisis kapasitas hantar arus kabel, penurunan tegangan pada kondisi operasi normal dan saat starting motor, serta penerapan faktor koreksi sesuai metode instalasi dan kondisi lingkungan. Hasil perhitungan ini digunakan sebagai dasar pemilihan ukuran kabel yang aman, andal, dan memenuhi standar serta referensi teknis yang berlaku, sekaligus sebagai acuan dalam proses pengadaan dan pelaksanaan konstruksi.

1.4 Istilah

Dokumen ini berisi istilah-istilah yang memiliki definisi tercantum sebagai berikut:

Tabel 1. 1 Daftar Istilah

Perusahaan	Perusahaan yang dimaksud adalah PT PERTAMINA HULU ROKAN selaku Pemilik dan Pengguna Fasilitas akhir.
Kontraktor	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan pengadaan dan konstruksi yang akan bertanggung jawab kepada PERUSAHAAN.
Vendor	Pihak lain yang akan menyediakan sesuatu untuk kebutuhan proyek, selain yang disediakan oleh PERUSAHAAN dan KONTRAKTOR, disebut VENDOR. Tanggung jawabnya langsung kepada KONTRAKTOR baik terhadap PO maupun bentuk perjanjian lain antara VENDOR dan KONTRAKTOR.
SKK MIGAS	Suatu badan dari Pemerintah Indonesia yang bertanggung jawab terhadap proses persetujuan dan izin untuk semua fasilitas di lingkungan minyak dan gas.
Perencana	Pihak yang dipilih oleh PERUSAHAAN sebagai pelaksana pekerjaan Jasa DED Engineering Services. Dalam hal ini disebut PERENCANA, yang akan bertanggung jawab kepada PERUSAHAAN sebagaimana dirinci dalam kontrak, scope of work, atau work order.
Inspektor	Perwakilan dari PERUSAHAAN dan/atau seseorang yang ditunjuk oleh PERUSAHAAN, yang bertanggung jawab sebagai INSPEKTOR atas pekerjaan yang dilakukan oleh KONTRAKTOR.SINGKATAN

2. KODE, STANDAR, DAN REFERENSI

Tabel 2. 1 International Code and Standard

No	Referensi	Deskripsi
1	IEEE 399	IEEE Recommended Practice for Industrial and Commercial Power Systems Analysis
2	NFPA 70 / NEC	National Electrical Code
3	NEMA WC70	Power Cables Rated 2000 Volts or Less for the Distribution of Electrical Energy (ICEA S-95-658)
4	ICEA S-73-532	Standard for Control, Thermocouple, Extension and Instrumentation Cable
5	ICEA S-95-658	Power Cables Rated 2000 Volts or Less for the Distribution of Electrical Energy

DOKUMEN PROYEK

No.	No. Dokumen		Nama Dokumen
1	TMBP-01-PR-DET-D107	-	Single Line Diagram Switchgear
2	TMBP-01-PI-DET-D108	-	Single Line Diagram
3	TMBP-01-PR-CAL-D101	-	Electrical Demand Load Analysis and Main Equiment Sizing

3. DAFTAR SINGKATAN

Beberapa singkatan yang paling umum digunakan tercantum dalam daftar berikut:

Tabel 3. 1 Daftar Singkatan

No	Singkatan	Deskripsi
1	AC	Alternating Current
2	AWG	American Wire Gauge
3	CRC	Cable Rated Current
4	HP	Horse Power
5	In	Nominal Current
6	kA	Kilo Ampere
7	kV	Kilo Volt
8	kVA	Kilo Volt Ampere
9	LV	Low Voltage

No	Singkatan	Deskripsi
10	MC	Metal Clad
11	MV	Medium Voltage
12	MCC	Motor Control Center
13	XHHW	Cross-Linked High Heat Water Resistant Insulated Wire

4. KRITERIA DAN ASUMSI DESAIN

4.1 Kriteria dan Asumsi

Kriteria dasar untuk perhitungan penentuan kabel adalah sebagai berikut :

a. Kapasitas arus yang diperbolehkan dalam kondisi *steady state*
b. *Voltage drop* (Tegangan jatuh) yang diperbolehkan dalam kondisi steady state.
c. *Voltage drop* (Tegangan jatuh) yang diperbolehkan saat starting motor.

4.2 Persyaratan Suhu dan Kondisi Operasi Normal

a. Kabel harus memiliki rating untuk suhu lingkungan 40°C (104°F) di udara.
b. Material konduktor harus berupa tembaga dengan isolasi *Cross-Linked Polyethylene High Heat Resistant* (XLPE) yang memiliki rating suhu 90°C (194°F), dengan lapisan luar dan dalam berbahan PVC, neoprene, atau hypalon.
c. Suhu maksimum konduktor adalah 90°C (194°F) baik pada lokasi basah maupun kering.

4.3 Voltage Volt (Penurunan Tegangan yang Diperbolehkan

Tegangan jatuh yang diperbolehkan berdasarkan *based practice* adalah sebagai berikut:

Tabel 4. 1 Penurunan Tegangan yang Diperbolehkan

During Normal Operation	**Voltage Drop**
Main feeders	2 %
Motor feeders	3 %
Transformer feeders	3 %
Panel board feeders	2 %
Lighting and Receptacle branch feeders	2 %
During Starting of Motors	**Voltage Drop**
380 Volt Bus	15 %

5. METODE PERHITUNGAN

Kriteria dasar untuk perhitungan penentuan kabel adalah sebagai berikut :

a. Kapasitas arus yang diperbolehkan dalam kondisi *steady state*
b. *Voltage drop* (Tegangan jatuh) yang diperbolehkan dalam kondisi *steady state*.
c. *Voltage drop* (Tegangan jatuh) yang diperbolehkan saat starting motor.

5.1 Kapasitas Arus Kabel

Rumus kapsitas arus kabel sebagai berikut:

$$I_n \times F \geq I_{FL} \times MF$$

$$I_{FL} = \frac{P}{\sqrt{3}\ x\ V_{L-L} x\ pf\ x\ eff}\ ,$$ *for 3-phase*

$$I_{FL} = \frac{P}{V_{L-N} x\ pf\ x\ eff}\ ,$$ *for 1-phase and DC system*

Dimana

I_n	:	Rating kapasitas arus kontinu kabel (Ampere)
I_{FL}	:	Arus beban penuh (Ampere)
P	:	Daya (kW)
V_{L-L}	:	Tegangan line-to-line (Volt)
V_{L-N}	:	Tegangan line-to-neutral (Volt)
pf	:	Faktor daya (%)
eff	:	Efisiensi (%)
F	:	Faktor derating
MF	:	Faktor pengali

Faktor pengali yang harus diperhitungkan dalam perhitungan ukuran kabel adalah:

- Faktor pengali untuk *incoming feede*r = 1.00 dari rating *ampere breaker incoming*.
- Faktor pengali untuk motor = 1.25 dari arus beban penuh, sesuai NEC Ed. 2023, article 430-22.
- Faktor pengali untuk *feeder* = 1.25 dari arus beban penuh, sesuai NEC Ed. 2023, article 215-2.

Cable Ampacity

Sebagai data awal untuk *ampacity* kabel, resistansi dan reaktansi mengacu pada rating *ampacity* berdasarkan [Ref. 2] NEC Ed. 2023. Data ini akan diperbarui setelah pemilihan pabrikan kabel final.

Faktor Derating (Df) untuk Kabel yang Diletakkan di Tray (Di Atas Tanah)

Rating *ampacity* dikurangi berdasarkan metode instalasi *actual*. *Factor* derating yang digunakan adalah :

1. Faktor koreksi temperatur (Kt) Berdasarkan [Ref.2] NEC Ed. 2023, faktor koreksi sebesar 0,91 diterapkan.

2. Faktor koreksi instalasi pengelompokan (Kg) Mengacu [ref.2] NEC Ed. 2023, dengan asumsi maksimum 6 kabel horizontal dalam satu lapisan, faktor koreksi adalah 0,70.

   Total Faktor Derating adalah sebagai berikut :

   Df = Kt x Kg
   = 0.91 x 0.56
   = **0.637**

5.2 Voltage Drop on Running Load

Rumus untuk menentukan tegangan jatuh pada sistem AC tiga fasa (*line to line*) adalah:

$$V_{D(L-L)}\ \% = \sqrt{3}\, x\, I \times \left(R\cos\phi + X\sin\phi\right) \times \frac{L}{1000} \times \frac{100\%}{V_{L-L}} \times \frac{1}{n}$$

Rumus untuk menentukan tegangan jatuh pada sistem AC satu fasa (*line to neutral*) adalah:

Rumus untuk menentukan tegangan jatuh pada sistem DC adalah:

$$V_{D(L-N)}\% = 2\, x\, I \times \left(R\cos\phi + X\sin\phi\right) \times \frac{L}{1000} \times \frac{100\%}{V_{L-N}} \times \frac{1}{n}$$

$$V_D\% = 2\, x\, I \times R_{DC} \times \frac{L}{1000} \times \frac{100\%}{V_{DC}} \times \frac{1}{n}$$

Dimana :

$V_{D\%}$	:	Voltage Drop (Tegangan jatuh) (%)
V_{L-L}	:	Tegangan line-to-line (Volt)
V_{L-N}	:	Tegangan line-to-neutral (Volt)
I	:	Arus beban (Ampere)
R	:	Resistansi kabel (Ω/km)
X	:	Reaktansi kabel (Ω/km)
φ	:	*Power Factor Angle*
L	:	Panjang kabel (m)
n	:	Jumlah kabel yang berjalan paralel

5.3 Voltage Drop at Motor Starting

Tegangan jatuh sistem AC saat starting motor diberikan oleh rumus berikut: Rumus tegangan jatuh saat starting motor pada sistem AC tiga fasa:

$$V_{dstart}\ \% = \sqrt{3}\, xIs \times \left(R\cos\phi s + X\sin\phi s\right) \times \frac{L}{1000} \times \frac{100\%}{V_{L-L}} \times \frac{1}{n}$$

Rumus tegangan jatuh saat starting motor pada sistem AC satu fasa:

$$V_{dstart}\% = 2x\,Is \times \left(R\cos\phi s + X\sin\phi s\right) \times \frac{L}{1000} \times \frac{100\%}{V_{L-N}} \times \frac{1}{n}$$

Dimana:

V*d start*	:	*Voltage Drop* (Tegangan jatuh) saat starting motor (%)
V_{L-L}	:	Tegangan *line-to-line* (Volt)
V_{L-N}	:	Tegangan *line-to-neutral* (Volt)
Is	:	Arus starting motor (Ampere)
φs	:	Motor Starting *Power Factor Angle*
L	:	Panjang kabel (m)
n	:	Jumlah kabel yang berjalan paralel
R	:	Resistansi kabel (Ω/km)
X	:	Reaktansi kabel (Ω/km)

LAMPIRAN

ELECTRICAL CABLE SIZING RESULT

skkmigas PERTAMINA EP

DED PENGEMBANGAN FASILITAS PRODUKSI
TANJUNG MIRING BARAT & KARANGAN TIMUR

POG

JOB NO :
DOC NO : TMBP-SP001-EL-CAL-D103
DATE : 9-Jan-2026
REV : A

ATTACHMENT - 1
CABLE SIZING CALCULATION

No.	CABLE TAG	FROM: EQUIPMENT TAG. NO.	FROM: EQUIPMENT DESCRIPTION	FROM: EQUIPMENT LOCATION (NOTE-2)	TO: EQUIPMENT TAG. NO.	TO: EQUIPMENT DESCRIPTION	TO: EQUIPMENT LOCATION (NOTE-2)	System Voltage (Vs) (Volt)	Load Type (F/M)	Starter Type	P (kW)	P (kVA)	Efficiency	PF	Full Load Current IFL (In Amps) = P(1/3 x Vs x cos Ø x Eff)	Cable Type	Cable Run (n)	Cable Size (mm2)	In Air / In Ground	Cable Ampacity (A)	Group Derating Factor	Total Ampacity =(n*A*d2)	Total Ampacity > Full Load Current	Cable Type	No. of Parallel Runs (N)	Total Cable Size (mm2)	Estimated Cable (in meters) (L)	Cable Voltage Drop Vd: Formula per Cl. 4.2 for AC & DC System (Vd)	Cable Voltage Drop Vd: Steady State: %Vd (Vd/Vs)	Cable Voltage Drop Vd: Permissible Voltage Drop (%)	Cable Voltage Drop Vd: Volt. Drop Limit Acceptable (Yes/No)	Required Minimum Cable Size based on Isc Requirement: CB rating selected (A)	I	S2 x k2	I2 x t	Cable Size > Min. Cable Size (Isc)	Final Cable Selected: No. of Parallel Runs (n)	No. of Cores	Cross-section (sqmm)	OD (mm)	Total Cable Length (m) =n.X.L	Total Cable Weight (kg) =(WL).X.L	Remarks
A.	**INCOMING SWITCHGEAR GEG & DEG**																																										
	GEG																																										
	P-GEG-001-001-004	GEG-001	Gas Engine Generator #2	D5	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	400	F		485	570.59	0.950	0.850	866.94	N2XRGbY-1-35	3	300	Ground	732	0.63	1383	Yes	N2XRGbY-1-35	3	900	90	5.241	1.310	2.00	Yes	1600	0.3	1.64E+09	7.68E+09	Yes	3 x 4 Phase	1C	300	37.97	1104	4146.624	3 parallel cables run per phase
	P-GEG-002-001-005	GEG-002	Gas Engine Generator #2	D5	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	400	F		485	570.59	0.950	0.850	866.94	N2XRGbY-1-35	3	300	Ground	732	0.63	1383	Yes	N2XRGbY-1-35	3	900	90	5.241	1.310	2.00	Yes	1600	0.3	1.64E+09	7.68E+09	Yes	3 x 4 Phase	1C	300	37.97	1104	4077.042	3 parallel cables run per phase
	DEG																																										
	P-DEG-001-001-004	DEG-001	Diesel Engine Generator #2	D5	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	400	F		485	570.59	0.950	0.850	866.94	N2XRGbY-1-35	3	300	Ground	732	0.63	1383	Yes	N2XRGbY-1-35	3	900	90	5.241	1.310	2.00	Yes	1600	0.3	1.64E+09	7.68E+09	Yes	3 x 4 Phase	1C	300	37.97	1104	4146.624	3 parallel cables run per phase
	P-DEG-002-001-005	DEG-002	Diesel Engine Generator #2	D5	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	400	F		485	570.59	0.950	0.850	866.94	N2XRGbY-1-35	3	300	Ground	732	0.63	1383	Yes	N2XRGbY-1-35	3	900	90	5.241	1.310	2.00	Yes	1600	0.3	1.64E+09	7.68E+09	Yes	3 x 4 Phase	1C	300	37.97	1104	4077.042	3 parallel cables run per phase
B.	**OUTGOING SWG/MCC BUS A**																																										
	Test Separator System																																										
1	P-S02A-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S02A	Test Pump TMS - A	D5	400	M	DOL	2.2	2.70	0.843	0.810	4.62	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	114	4.417	1.104	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	455	90.738	
2	P-S03A-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S03A	Test Pump KRG - A	D5	400	M	DOL	3	3.68	0.850	0.820	6.18	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	123	6.417	1.604	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	492	98.071	
3	P-S03B-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S03B	Test Pump KRG - B	D5	400	M	DOL	3	3.68	0.850	0.820	6.18	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	139	6.177	1.544	5.00	Yes	10	0.3	6.00E+05	3.00E+01	No	1	3C	4	19.72	472	94.401	
	Production Water																																										
	Water Injection System																																										
	K.O Drum System																																										
13	P-O06A-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-O06A	KOD Pump - A	D6	400	M	DOL	0.55	0.85	0.655	0.650	1.86	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	146	1.602	0.456	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	580	115.485	
14	P-O06B-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-O06B	KOD Pump - B	D6	400	M	DOL	0.55	0.85	0.655	0.650	1.86	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	148	1.665	0.466	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	593	118.235	
15	P-S21-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S21	Test Sump Pump	D6	400	M	DOL	0.37	0.67	0.650	0.650	1.25	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	140	1.187	0.297	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	561	111.819	
	Closed Drain System																																										
16	P-S07A-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S07A	Closed Drain Pump - A	D6	400	M	DOL	1.1	1.48	0.814	0.780	2.57	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	153	3.080	0.770	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	603	120.097	
17	P-S07B-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S07B	Closed Drain Pump - B	D6	400	M	DOL	1.1	1.48	0.814	0.780	2.57	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	156	3.172	0.793	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	630	125.557	
18	P-S18-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S18	Closed Drain Sump Pump	D6	400	M	DOL	0.55	0.76	0.781	0.725	1.40	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	160	1.680	0.420	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	639	127.400	
	Oil Pit																																										
19	P-O19-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-O19	Water Sump Pump	C4	400	M	DOL	1.5	1.85	0.828	0.810	3.23	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	233	6.288	1.572	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	934	186.060	
	Storage Tank KRG																																										
22	Storage Tank Sump Pump - KRG	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S40	Storage Tank Sump Pump - KRG	C4	400	M	DOL	0.75	1.02	0.796	0.730	1.85	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	200	2.801	0.700	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	800	159.600	
25	P-S88-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S88	Biocide Chemical Injection - Package	C3	400	M	DOL	1.48	1.85	0.800	0.800	2.60	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	209	2.860	0.625	5.00	Yes	25	0.3	2.04E+06	1.88E+02	Yes	1	4C	10	25.16	1078	388.313	
27	P-S89-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	P-S89	Biocide Chemical Transfer Pump	C3	400	M	DOL	1.1	1.48	0.814	0.780	2.57	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	209	2.486	1.271	5.00	Yes	25	0.3	2.04E+06	1.88E+02	Yes	1	4C	10	25.16	1078	217.222	
	Electrical Feeder																																										
28	P-UPS-001A-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	UPS-001A	AC UPS System Normal	E6	400	F		26.5	30.00	0.950	0.850	45.58	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	43	6.943	1.736	5.00	Yes	63	0.3	2.04E+06	1.19E+03	Yes	1	4C	10	25.16	175	62.561	
29	P-NLP-001-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	NLP-001	Normal Lighting Panel and Small Power DB	E6	400	F		10.37	12.20	0.950	0.850	18.54	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	45	2.899	0.725	5.00	Yes	25	0.3	2.04E+06	1.88E+02	Yes	1	4C	10	25.16	178	64.221	
30	P-CHC-002-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	CHC-002	Crane for Electric Chain Hoist for Test & Transfer Pump Shelter	D6	400	F		2	2.35	0.950	0.850	3.58	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	123	1.546	0.386	5.00	Yes	10	0.3	2.04E+06	3.00E+01	Yes	1	4C	10	25.16	492	177.561	
33	P-CHC-005-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	CHC-005	Crane for Electric Chain Hoist for Instrument Air Package Shelter	D6	400	F		9.5	11.18	0.950	0.850	16.98	N2XRGbY-4-05	1	10	Ground	85	0.63	54	Yes	N2XRGbY-4-05	1	10	117	6.999	1.750	5.00	Yes	25	0.3	2.04E+06	1.88E+02	Yes	1	4C	10	25.16	469	169.264	
34	P-CHC-006-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	CHC-006	Crane for Electric Chain Hoist for Fire Water Pump Shelter	C4	400	F		2	2.35	0.950	0.850	3.58	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	322	4.045	1.011	5.00	Yes	10	0.3	2.04E+06	3.00E+01	Yes	1	4C	10	25.16	1288	464.646	
35	P-CHC-007-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	CHC-007	Crane for Electric Chain Hoist for City Pit Shelter Shelter	C3	400	F		2	2.35	0.950	0.850	3.58	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	237	2.976	0.744	5.00	Yes	10	0.3	2.04E+06	3.00E+01	Yes	1	4C	10	25.16	948	341.847	
36	P-CHC-008-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	CHC-008	Crane for Electric Chain Hoist for Gross Transfer Pump Shelter	C5	400	F		2	2.35	0.950	0.850	3.58	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	224	2.817	0.704	5.00	Yes	10	0.3	2.04E+06	3.00E+01	Yes	1	4C	10	25.16	897	323.593	
37	P-WO-002-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	WO-001	Welding Outlet for GEG & EGG Shelter	D6	400	F		26	29.41	0.950	0.850	44.69	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	89	14.048	3.512	5.00	Yes	80	0.3	2.04E+06	7.50E+02	Yes	1	4C	10	25.16	398	139.106	
38	P-WO-003-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	WO-001	Welding Outlet for Instrument/Fuel Building	D6	400	F		25	29.41	0.950	0.850	44.69	N2XRGbY-4-07	1	25	Air	138	0.66	91	Yes	N2XRGbY-4-07	1	25	183	10.726	2.681	5.00	Yes	80	0.3	1.28E+07	7.50E+02	Yes	1	4C	25	30.74	549	429.856	
39	P-WO-004-001	SWG/MCC-001A	0.4 kV Low Voltage Switchgear & MCC Bus A	E6	WO-004	Welding Outlet for KOD Drum Shelter	C5	400	F		25	29.41	0.950	0.850	44.69	N2XRGbY-4-07	1	25	Air	138	0.66	91	Yes	N2XRGbY-4-07	1	25	148	9.513	2.378	5.00	Yes	50	0.3	1.28E+07	7.50E+02	Yes	1	4C	25	30.74	593	345.507	
C.	**OUTGOING SWG/MCC BUS B**																																										
	Gross Oil Transfer System																																										
1	P-S0-P-001A-001-002	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	S0-P-001A	Panel Sdt Starter for Gross Transfer Pump - A	E6	400	M	SS	90	102.86	0.942	0.875	157.61	N2XRGbY-3-11	1	95	Air	279	0.66	182	Yes	N2XRGbY-3-11	1	95	25	1.365	0.341	5.00	Yes	400	0.3	1.85E+08	4.80E+04	Yes	1	3C	95	43.71	85	103.180	
2	P-S0-A-001-002	SWG/MCC-001B	Panel Sdt for Gross Transfer Pumps - A	E6	P-S01A	Process Control Transfer Pump - A	C3	400	M	SS	90	102.86	0.942	0.875	157.61	N2XRGbY-3-11	1	95	Air	279	0.66	182	Yes	N2XRGbY-3-11	1	95	270	14.912	3.728	5.00	Yes	400	0.3	1.85E+08	4.80E+04	Yes	1	3C	95	43.71	874	1127.000	
3	P-S0-P-001B-001-002	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	S0-P-001B	Panel Sdt Starter for Gross Transfer Pump - B	E6	400	M	SS	90	102.86	0.942	0.875	157.61	N2XRGbY-3-11	1	95	Air	279	0.66	182	Yes	N2XRGbY-3-11	1	95	25	1.365	0.341	5.00	Yes	400	0.3	1.85E+08	4.80E+04	Yes	1	3C	95	43.71	85	103.180	
4	P-S0-B-001	SWG/MCC-001B	Panel Sdt for Gross Transfer Pumps - B	E6	P-S01B	Process Control Transfer Pump - B	C3	400	M	SS	90	102.86	0.942	0.875	157.61	N2XRGbY-3-11	1	95	Air	279	0.66	182	Yes	N2XRGbY-3-11	1	95	270	14.919	3.730	5.00	Yes	400	0.3	1.85E+08	4.80E+04	Yes	1	3C	95	43.71	874	1127.000	
5	P-S01-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	P-S03	Gross Transfer Pump	C5	400	M	DOL	0.37	0.67	0.650	0.650	1.25	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	242	2.043	0.511	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	968	192.476	
	IWGA System																																										
6	P-C-001A-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	C-001A	Instrument Air Compressor - A	E7	400	F		75	88.24	0.950	0.850	134.06	N2XRGbY-4-11	1	95	Ground	280	0.63	176	Yes	N2XRGbY-4-11	1	95	104	5.932	1.483	5.00	Yes	160	0.3	1.85E+08	7.68E+03	Yes	1	4C	95	48.72	414	689.621	
7	P-C-001B-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	C-001B	Instrument Air Compressor - B	E7	400	F		75	88.24	0.950	0.850	134.06	N2XRGbY-4-11	1	95	Ground	280	0.63	176	Yes	N2XRGbY-4-11	1	95	139	6.261	1.565	5.00	Yes	160	0.3	1.85E+08	7.68E+03	Yes	1	4C	95	48.72	420	737.935	
	Potable Water System																																										
8	P-P-008A-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	P-008A	Deep Well Pump - A	E4	400	M	DOL	0.75	1.02	0.796	0.730	1.85	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	130	2.201	0.550	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	538	106.153	
9	P-P-008B-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	P-008B	Deep Well Pump - B	E4	400	M	DOL	0.75	1.02	0.796	0.730	1.85	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	147	2.069	0.517	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	586	117.318	
10	P-P-016A-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	P-016A	Booster Pump - A	E7	400	M	DOL	0.37	0.57	0.650	0.650	1.25	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	178	1.508	0.377	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	713	142.065	
11	P-P-016B-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	P-016B	Booster Pump - B	E7	400	M	DOL	0.37	0.57	0.650	0.650	1.25	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	175	1.479	0.370	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	689	138.316	
12	P-P-017A-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	P-017A	Water Distribution Pump - A	D7	400	M	DOL	0.37	0.57	0.650	0.650	1.25	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	176	1.488	0.372	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	704	140.232	
13	P-P-017B-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	P-017B	Water Distribution Pump - B	D7	400	M	DOL	0.37	0.57	0.650	0.650	1.25	N2XRGbY-3-03	1	4	Air	51	0.66	34	Yes	N2XRGbY-3-03	1	4	173	1.459	0.365	5.00	Yes	10	0.3	3.27E+05	3.00E+01	Yes	1	3C	4	19.72	691	137.480	
	Fire Water System																																										
14	P-P-001-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	P-001	Jockey Pump	C2	400	F		2.2	2.59	0.850	0.850	3.85	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	397	1.243	0.311	5.00	Yes	10	0.3	2.04E+06	3.00E+01	Yes	1	4C	10	25.16	1228	440.033	
	Electrical Feeder																																										
15	P-UPS-001B-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	UPS-001B	AC UPS Redundant	E6	400	F		26.5	30.00	0.950	0.850	45.58	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	42	6.759	1.690	5.00	Yes	50	0.3	2.04E+06	7.50E+02	Yes	1	4C	10	25.16	169	60.907	
16	P-UPS-001C-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	UPS-001C	AC UPS (By Pass)	E6	400	F		26.5	30.00	0.950	0.850	45.58	N2XRGbY-4-05	1	10	Air	85	0.66	56	Yes	N2XRGbY-4-05	1	10	43	6.943	1.736	5.00	Yes	50	0.3	2.04E+06	7.50E+02	Yes	1	4C	10	25.16	175	62.561	
17	P-ELP-001-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	ELP-001	Emergency Lighting Panel and Small Power DB	E6	400	F		54.25	63.82	0.950	0.850	96.97	N2XRGbY-4-08	1	35	Air	164	0.66	106	Yes	N2XRGbY-4-08	1	35	46	4.653	1.163	5.00	Yes	125	0.3	2.51E+07	4.80E+03	Yes	1	4C	35	34.5	183	141.667	
18	P-CHC-001-001	SWG/MCC-001B	0.4 kV Low Voltage Switchgear & MCC Bus B	E6	CHC-001	Overhead Crane for GEG Shelter	C3	400	F		9.5	11.18	0.950	0.850	16.98	N2XRGbY-4-05	1	10	Ground	85	0.63	54	Yes	N2XRGbY-4-05	1	10	139	5.338	1.335	5.00	Yes	25	0.3	2.04E+06	1.88E+02	Yes	1	4C	10	25.16	558	139.135	
D.	**OUTGOING AC UPS**																																										
1	P-BAT-UPS-001-001	DB-BAT-UPS-001	BCCB BOX BATTERY	E6	BAT-UPS-001	Battery UPS	E6	220	F		26.5	30.00	0.950	0.850	143.54	FRC-2-02	1	50	Air	229	0.66	151	Yes	FRC-2-02	1	50	30	3.436	1.796	5.00	Yes	160	0.3	5.11E+07	7.68E+03	Yes	1	2C	50	28.66	60	46.256	
2	P-DB-BAT-UPS-001-001	DB-BAT-UPS-001	BCCB BOX BATTERY	E6	DB-BAT-UPS-001	BCCB BOX BATTERY	E6	220	F		26.5	30.00	0.950	0.850	143.54	FRC-2-02	1	50	Air	229	0.66	151	Yes	FRC-2-02	1	50	30	3.436	1.796	5.00	Yes	160	0.3	5.11E+07	7.68E+03	Yes	1	2C	50	28.66	60	46.256	
3	P-TMBP-PCS-001-001	DB-UPS-001	Dist. Board UPS	E6	TMBP-PCS-001	Process Control System (PCS)	E6	230	F		1.13	1.33	0.950	0.850	6.08	FRC-2-02	1	2.5	Air	47	0.66	31	Yes	FRC-2-02	1	2.5	25	2.464	1.070	5.00	Yes	10	0.3	1.28E+05	3.00E+01	Yes	1	3C	2.5	15.34	50	8.050	Redundant Feeder
4	P-TMBP-PCS-001-002	DB-UPS-001	Dist. Board UPS	E6	TMBP-PCS-001	Process Control System (PCS)	E6	230	F		1.13	1.33	0.950	0.850	6.08	FRC-2-02	1	2.5	Air	47	0.66	31	Yes	FRC-2-02	1	2.5	25	2.464	1.070	5.00	Yes	10	0.3	1.28E+05	3.00E+01	Yes	1	3C	2.5	15.34	50	8.050	Redundant Feeder
5	P-TMBP-ESS-002-001	DB-UPS-001	Dist. Board UPS	E6	TMBP-ESS-002	Emergency Shutdown System (ESD) / Fire Gas System (FGS)	E6	230	F		1.13	1.33	0.950	0.850	6.08	FRC-2-02	1	2.5	Air	47	0.66	31	Yes	FRC-2-02	1	2.5	25	2.464	1.070	5.00	Yes	10	0.3	1.28E+05	3.00E+01	Yes	1	3C	2.5	15.34	50	8.050	Redundant Feeder
6	P-TMBP-ESS-002-002	DB-UPS-001	Dist. Board UPS	E6	TMBP-ESS-002	Emergency Shutdown System (ESD) / Fire Gas System (FGS)	E6	230	F		1.13	1.33	0.950	0.850	6.08	FRC-2-02	1	2.5	Air	47	0.66	31	Yes	FRC-2-02	1	2.5	25	2.464	1.070

LAMPIRAN - 2
CALCULATION SAMPLE - 1 (400 VAC FEEDER)

Cable Tag		=	P-GEG-001-001~004		
Cable Route:	- From	=	SWG/MCC-001A ,	0.4 kV Low Voltage Switchgear & MCC Bus A	
	- To	=	GEG-001 ,	Gas Engine Generator #1	
Installation Method		=	Ground		
Load Type		=	Feeder		
Voltage, Vn		=	400	V AC	
Equipment Rating, P		=	485	kW	
Power Factor, Cos φ		=	0.85	=> Sin φ =	0.53
Efficiency, Eff		=	0.95		
Total Derating Factor, K		=	0.63		
Temp. Correction Factor, K1a		=	1		
Group Derating Factor, K2a		=	0.42		
Soil Thermal Resistivity Factor, K3		=	1.5		
Approx. Cable Length, L		=	92	m	
Proposed Cable Size, S		=	1C	# 300 mm2	
No. of Cable Running in Parallel, n		=	3		
Cable Ampacity, IN		=	732.00	A	
Cable Resistance, R		=	0.08	Ω / km	
Cable Reactance, X		=	0.087	Ω / km	
Allowable Voltage Drop S/State, Vdn Max		=	1	%	
Allowable Voltage Drop Start, Vdnst Max		=	N/A		
Short Circuit Clearing Time, t		=	0.3	sec	
Constant Factor, k		=	143		

CABLE SIZING CALCULATION :

1.) Full Load Current

i) Full Load Current: ***IFL***

$$= \frac{P \times 1000}{\sqrt{3} \times Vn \times \cos\varphi \times Eff} = \frac{485000}{\sqrt{3} \times 400 \times 0.85 \times 0.95}$$

= **866.94** **A**

ii) Current Carrying Capacity: ***IZ*** = ***IN x K x n***

= 732 x 0.63 x 3 A

= **1383.48** **A**

Min. Design of Load Current for Cable:

IZ ≥ ***IFL*** => ACCEPTABLE

1383 A ≥ **866.94 A** **=> ACCEPTABLE**

2a.) Voltage Drop at Steady State

i) Calculated Voltage Drop at Steady State ***Vd***

$$= \frac{\sqrt{3} \times IFL \times [\,(R \cos\varphi) + (X \sin\varphi)\,] \times L)}{1000 \times n}$$

$$= \frac{\sqrt{3} \times 866.94 \times [\,(0.08 \times 0.85) + (0.087 \times 0.53)\,] \times 92)}{1000 \times 3}$$

= **5.254** **Volt**

ii) % Calculated Voltage Drop at Steady State ***Vdn***

$$= \frac{Vd}{Vn} \times 100\ \% = \frac{5.254}{400} \times 100\ \%$$

= **1.314** **%**

Vdn < ***Vdn Max*** => ACCEPTABLE

1.31 % < **2.00 %** **=> ACCEPTABLE**

3.) Short Circuit Thermal Withstand Capacity (Short Circuit Protection)

i) Cable Short Circuit Withstand Capacity ***S2k2*** = 300^{2a} x 143^{2a}

= **1840410000** **A2s**

ii) ***IFL*** = 866.92 A

Proposed CB rating = 1600 A

CB capacity let through energy ***I2t*** = **768000** **A2s**

I2t ≤ ***S2k2*** => ACCEPTABLE

768000 ≤ **1840410000** **=> ACCEPTABLE**

CONCLUSION

Hence, selected cable **3 x 1C(4) # 300** mm2 has met the minimum requirement of cable selection condition that Cable Cont. Current Carrying Capacity (IZ) ≥ Full Load Current (IFL) and Cable Voltage Drop on Steady State (Vdn) < Max Voltage Drop (VdnMax).

LAMPIRAN - 2

CALCULATION SAMPLE - 2 (400 VAC MOTOR STARTER (SS))

Cable Tag		=	P-SS-P-001A-001~002				
Cable Route:	- From	=	SWG/MCC-001B	,	0.4 kV Low Voltage Switchgear & MCC Bus B		
	- To	=	SS-P-001A	,	Panel Soft Starter for Gross Transfer Pump - A		
Installation Method		=	Air				
Load Type		=	Feeder				
Voltage, Vn		=	400	V AC			
Equipment Rating, P		=	90	kW			
Power Factor, Cos φ		=	0.875	=> Sin φ		=	0.48
Power Factor (at Starting), Cos φ		=	0.30	=> Sin φ		=	0.950
Efficiency, Eff		=	0.94				
Total Derating Factor, K		=	0.66				
Temp. Correction Factor, K1a		=	0.91				
Group Derating Factor, K2a		=	0.73				
Soil Thermal Resistivity Factor, K3		=	N/A				
Approx. Cable Length, L		=	20	m			
Proposed Cable Size, S		=	3C	#	95	mm2	
No. of Cable Running in Parallel, n		=	1				
Cable Ampacity, IN		=	320.00	A			
Cable Resistance, R		=	0.247	Ω / km			
Cable Reactance, X		=	0.07	Ω / km			
Allowable Voltage Drop S/State, Vdn Max		=	5	%			
Allowable Voltage Drop Start, Vdnst Max		=	N/A				
Short Circuit Clearing Time, t		=	0.3	sec			
Constant Factor, k		=	143				

CABLE SIZING CALCULATION :

1.) Full Load Current

i) Full Load Current: ***IFL*** = $\frac{P \times 1000}{\sqrt{3} \times Vn \times \cos\varphi \times Eff}$ = $\frac{90000}{\sqrt{3} \times 400 \times 0.875 \times 0.950}$

= **156.28** **A**

ii) Current Carrying Capacity: ***IZ*** = ***IN x K x n***

= 320 x 0.66 x 1 A

= **211.20** **A**

Min. Design of Load Current for Cable:

IZ ≥ ***IFL*** => ACCEPTABLE

211 A ≥ **156.28 A** **=> ACCEPTABLE**

2a.) Voltage Drop at Steady State

i) Calculated Voltage Drop at Steady State

$$Vd = \frac{\sqrt{3} \times IFL \times [\,(R \cos\varphi) + (X \sin\varphi)\,] \times L}{1000 \times n}$$

$$= \frac{\sqrt{3} \times 157.6 \times [\,(0.247 \times 0.875) + (0.07 \times 0.48)\,] \times 20}{1000 \times 1}$$

= **1.352** **Volt**

ii) % Calculated Voltage Drop at Steady State

$$Vdn = \frac{Vd}{Vn} \times 100\ \% = \frac{1.352}{400} \times 100\ \%$$

= **0.337973684** **%**

Vdn < ***Vdn Max*** => ACCEPTABLE

0.34 % < **5.00 %** **=> ACCEPTABLE**

2b.) Voltage Drop during Motor Starting

i) Starting Current ***Ist*** = 6.5 x ***IFL***

= 1.5 x *156.28* A

= **234.42** **A**

ii) Calculated Voltage Drop at Steady State

$$Vd = \frac{\sqrt{3} \times IST \times [\,(R \cos\varphi) + (X \sin\varphi)\,] \times L}{1000 \times 1}$$

$$= \frac{\sqrt{3} \times 236.4 \times [\,(0.247 \times 0.3) + (0.07 \times 0.95)\,] \times 20}{1000 \times 1}$$

= **2.028** **Volt**

iii) % Calculated Voltage Drop at Steady State

$$Vdn = \frac{Vd}{Vn} \times 100\ \% = \frac{2.028}{400} \times 100\ \%$$

= **0.507** **%**

Vdn < ***Vdn Max*** => ACCEPTABLE

0.51 % < **15.00 %** **=> ACCEPTABLE**

3.) Short Circuit Thermal Withstand Capacity (Short Circuit Protection)

i) Cable Short Circuit Withstand Capacity ***S2k2*** = 95^{2a} x 143^{2a}

= **184552225** **A2s**

ii) ***IFL*** = 157.60 A

Proposed CB rating = 250 A

CB capacity let through energy ***I2t*** = **18750** **A2s**

I2t ≤ ***S2k2*** => ACCEPTABLE

18750 ≤ **184552225** **=> ACCEPTABLE**

CONCLUSION

Hence, selected cable **1 x 3C # 95** mm2 has met the minimum requirement of cable selection condition that Cable Cont. Current Carrying Capacity (IZ) ≥ Full Load Current (IFL) and Cable Voltage Drop on Steady State (Vdn) < Max Voltage Drop (VdnMax).

LAMPIRAN - 2
CALCULATION SAMPLE - 3 (230 VAC FEEDER)

Cable Tag		=	P-TMBP-PCS-001-001			
Cable Route:	- From	=	DB-UPS-001 ,	Dist. Board UPS		
	- To	=	TMBP-PCS-001 ,	Process Control System (PCS)		
Installation Method		=	Air			
Load Type		=	Feeder			
Voltage, Vn		=	230	V AC		
Equipment Rating, P		=	1.13	kW		
Power Factor, Cos φ		=	0.85	=> Sin φ	=	0.53
Efficiency, Eff		=	0.95			
Total Derating Factor, K		=	0.66			
Temp. Correction Factor, K1a		=	0.91			
Group Derating Factor, K2a		=	0.73			
Soil Thermal Resistivity Factor, K3		=	N/A			
Approx. Cable Length, L		=	25	m		
Proposed Cable Size, S		=	2C	# 2.5 mm2		
No. of Cable Running in Parallel, n		=	1			
Cable Ampacity, IN		=	41.00	A		
Cable Resistance, R		=	9.449	Ω / km		
Cable Reactance, X		=	0.108	Ω / km		
Allowable Voltage Drop S/State, Vdn Max		=	2	%		
Allowable Voltage Drop Start, Vdnst Max		=	N/A			
Short Circuit Clearing Time, t		=	0.3	sec		
Constant Factor, k		=	143			

CABLE SIZING CALCULATION :

1.) Full Load Current

i) Full Load Current: ***IFL*** = $\frac{P \times 1000}{\sqrt{3} \times Vn \times \cos\varphi \times Eff}$ = $\frac{1130}{\sqrt{3} \times 230 \times 0.85 \times 0.95}$

= **3.51** **A**

ii) Current Carrying Capacity: ***IZ*** = ***IN x K x n***

= 41 x 0.6643 x 1 A

= **27.24** **A**

Min. Design of Load Current for Cable:

IZ ≥ ***IFL*** => ACCEPTABLE

27 A ≥ **3.51 A** => **ACCEPTABLE**

2a.) Voltage Drop at Steady State

i) Calculated Voltage Drop at Steady State ***Vd*** = $\frac{2 \times IFL \times [(R \cos\varphi) + (X \sin\varphi)] \times L}{1000 \times n}$

= $\frac{2 \times 3.51 \times [(9.449 \times 0.85) + (0.108 \times 0.53)] \times 25}{1000 \times 1}$

= **1.421** **Volt**

ii) % Calculated Voltage Drop at Steady State ***Vdn*** = $\frac{Vd}{Vn} \times 100\ \%$ = $\frac{1.421}{230} \times 100\ \%$

= **0.6177** **%**

Vdn < ***Vdn Max*** => ACCEPTABLE

1.42 % < **5.00 %** => **ACCEPTABLE**

3.) Short Circuit Thermal Withstand Capacity (Short Circuit Protection)

i) Cable Short Circuit Withstand Capacity ***S2k2*** = $2.5^{2a} \times 143^{2a}$

= **127806.25** **A2s**

ii) ***IFL*** = 6.08 A

Proposed CB rating = 10 A

CB capacity let through energy ***I2t*** = **30** **A2s**

I2t ≤ ***S2k2*** => ACCEPTABLE

30 ≤ **127806.25** => **ACCEPTABLE**

CONCLUSION

Hence, selected cable **1 x 2C # 2.5** mm2 has met the minimum requirement of cable selection condition that Cable Cont. Current Carrying Capacity (IZ) ≥ Full Load Current (IFL) and Cable Voltage Drop on Steady State (Vdn) < Max Voltage Drop (VdnMax).

LAMPIRAN - 2
CALCULATION SAMPLE - 4 (220 VDC FEEDER)

Cable Tag		=	P-DB-BAT-UPS-001-001			
Cable Route:	- From	=	UPS-001 ,	AC UPS		
	- To	=	DB-BAT-UPS-001 ,	MCCB BOX BATTERY		
Installation Method		=	Air			
Load Type		=	Feeder			
Voltage, Vn		=	220	V DC		
Equipment Rating, P		=	25.5	kW		
Power Factor, Cos φ		=	0.85	=> Sin φ	=	0.53
Efficiency, Eff		=	0.95			
Total Derating Factor, K		=	0.66			
Temp. Correction Factor, K1a		=	0.91			
Group Derating Factor, K2a		=	0.73			
Soil Thermal Resistivity Factor, K3		=	N/A			
Approx. Cable Length, L		=	30	m		
Proposed Cable Size, S		=	2C	# 50 mm2		
No. of Cable Running in Parallel, n		=	1			
Cable Ampacity, IN		=	237.00	A		
Cable Resistance, R		=	0.49	Ω / km		
Cable Reactance, X		=	0.07	Ω / km		
Allowable Voltage Drop S/State, Vdn Max		=	2	%		
Allowable Voltage Drop Start, Vdnst Max		=	N/A			
Short Circuit Clearing Time, t		=	0.3	sec		
Constant Factor, k		=	143			

CABLE SIZING CALCULATION :

1.) Full Load Current

i) Full Load Current: $IFL = \frac{P \times 1000}{Vn} = \frac{25500}{220}$

= **115.91** **A**

ii) Current Carrying Capacity: $IZ = IN \times K \times n$

= 237 x 0.66 x 1 A

= **156.42** **A**

Min. Design of Load Current for Cable:

IZ	≥	*IFL*	=>	ACCEPTABLE
156 A	≥	**115.91 A**	**=>**	**ACCEPTABLE**

2a.) Voltage Drop at Steady State

i) Calculated Voltage Drop at Steady State: $Vd = \frac{2 \times IFL \times R \times L}{1000 \times n}$

$= \frac{2 \times 115.91 \times 0.49 \times 30}{1000 \times 1}$

= **3.408** **Volt**

ii) % Calculated Voltage Drop at Steady State: $Vdn = \frac{Vd}{Vn} \times 100\ \% = \frac{3.408}{220} \times 100\ \%$

= **1.549** **%**

Vdn	<	*Vdn Max*	=>	ACCEPTABLE
1.55 %	<	**5.00 %**	**=>**	**ACCEPTABLE**

3.) Short Circuit Thermal Withstand Capacity (Short Circuit Protection)

i) Cable Short Circuit Withstand Capacity: $S2k2 = 50^{2a} \times 143^{2a}$

= **51122500** **A2s**

ii) *IFL* = 115.91 A

Proposed CB rating = 160 A

CB capacity let through energy: $I2t$ = **7680** **A2s**

I2t	≤	*S2k2*	=>	ACCEPTABLE
7680	≤	**51122500**	**=>**	**ACCEPTABLE**

CONCLUSION

Hence, selected cable **1 x 2C # 50** mm2 has met the minimum requirement of cable selection condition that Cable Cont. Current Carrying Capacity (IZ) ≥ Full Load Current (IFL) and Cable Voltage Drop on Steady State (Vdn) < Max Voltage Drop (VdnMax).